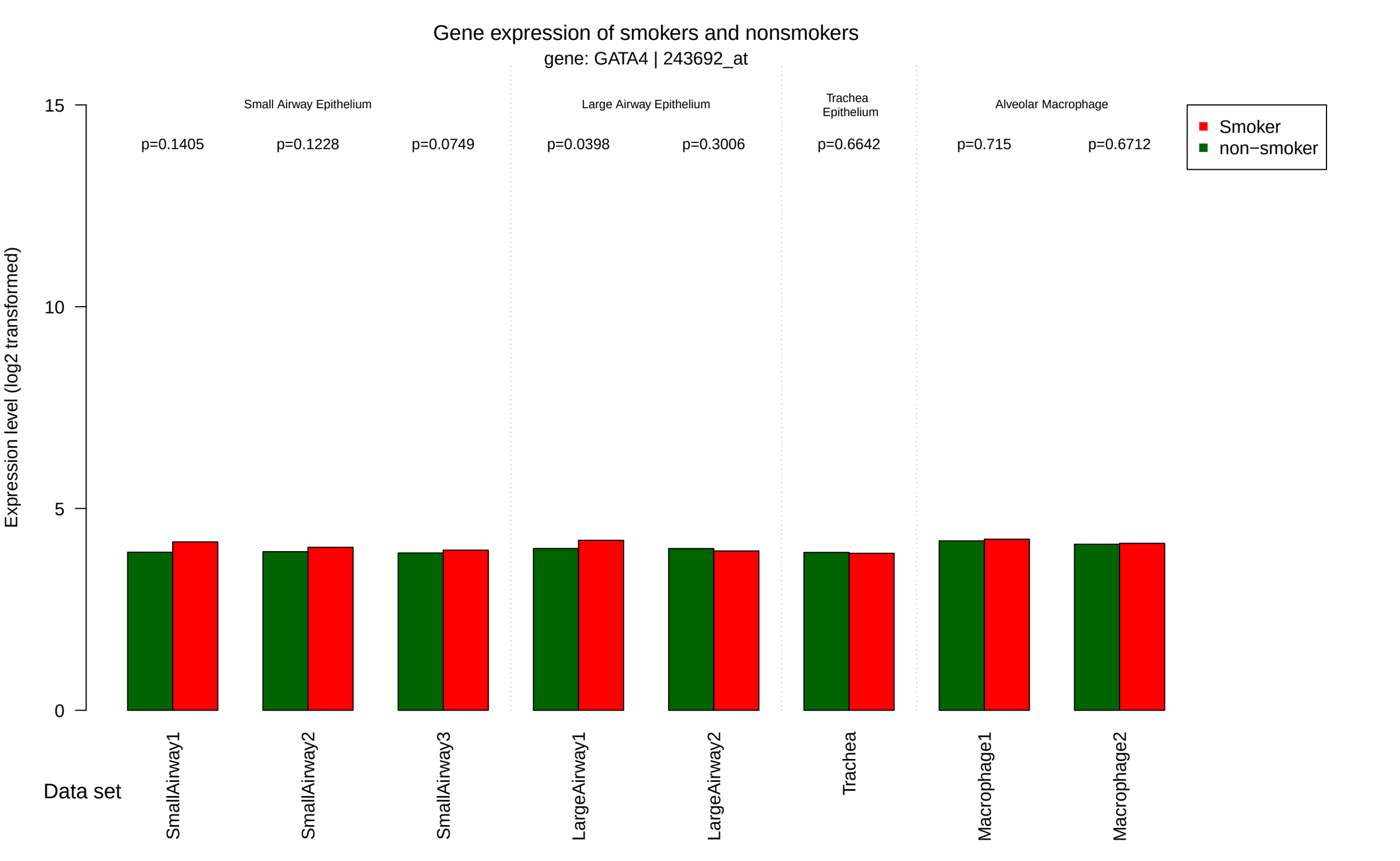
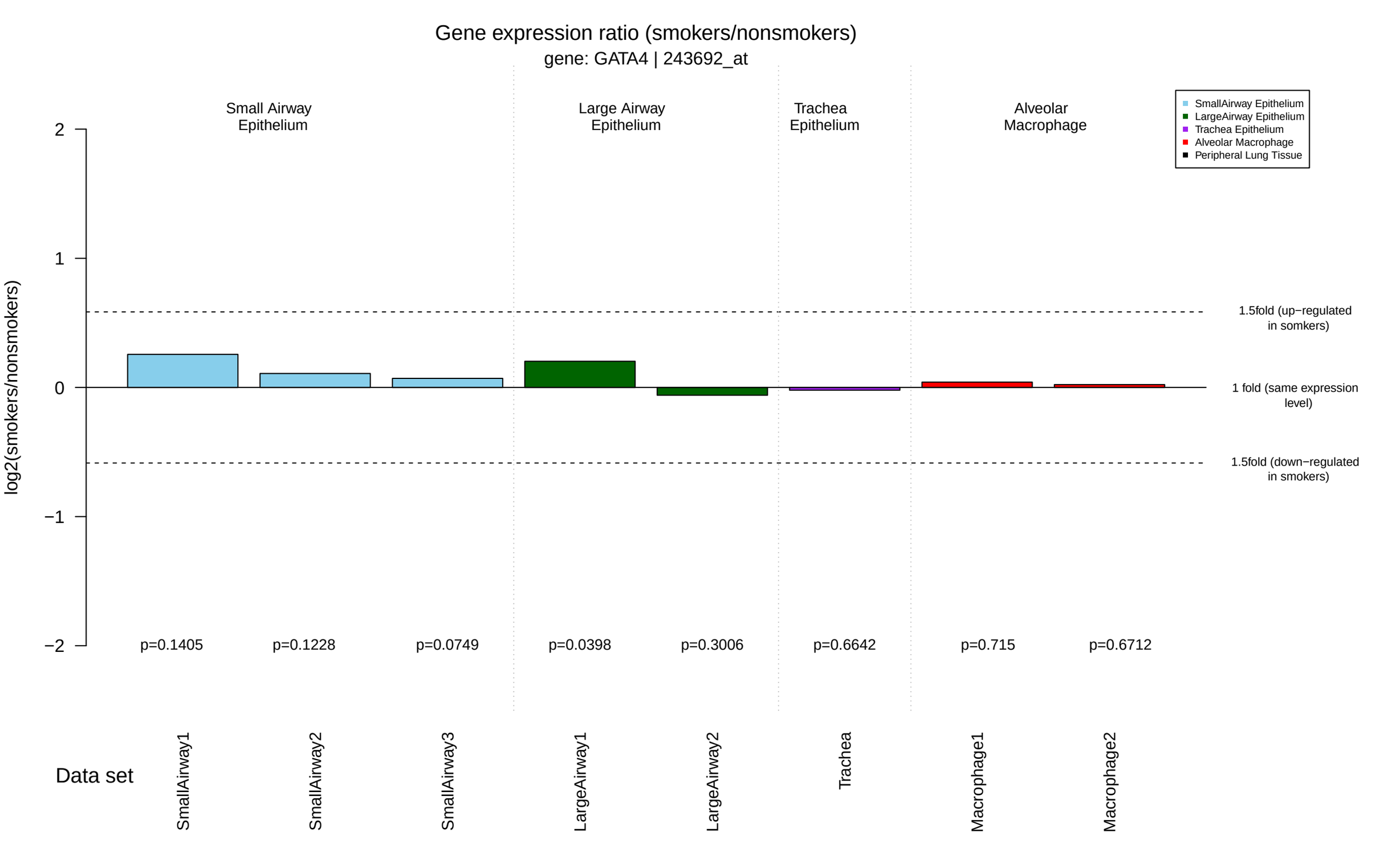
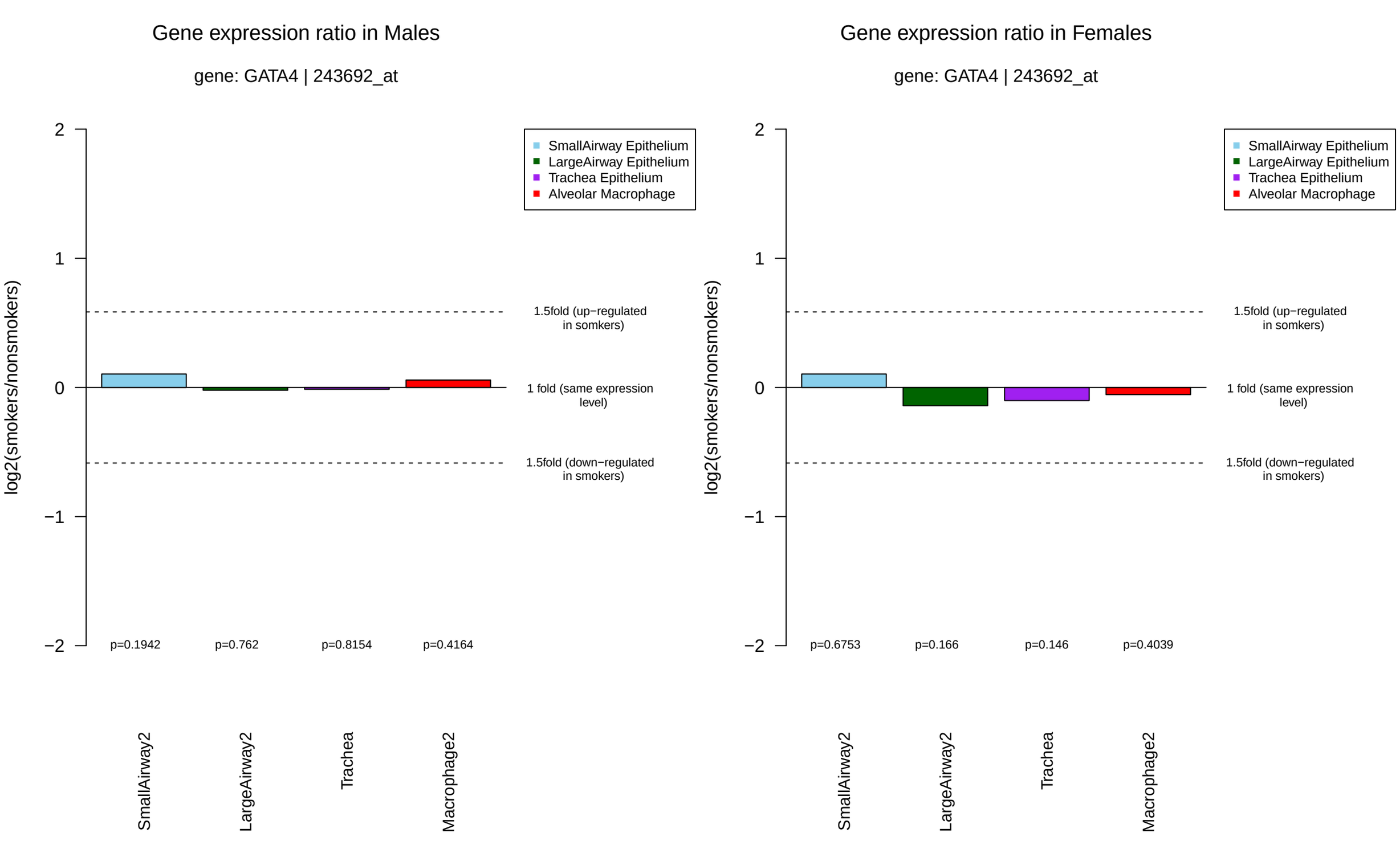
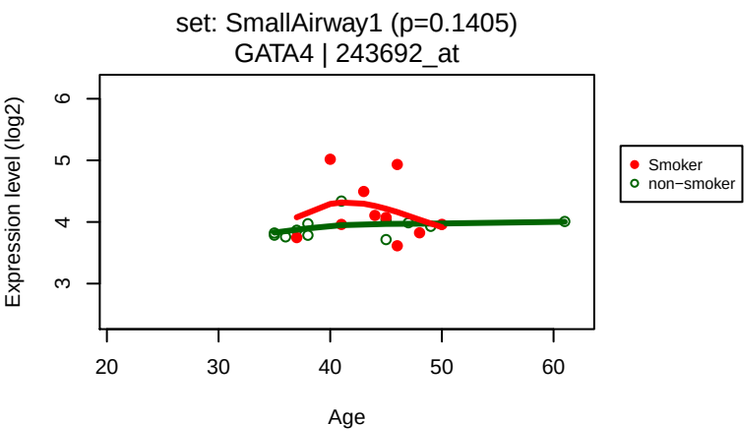
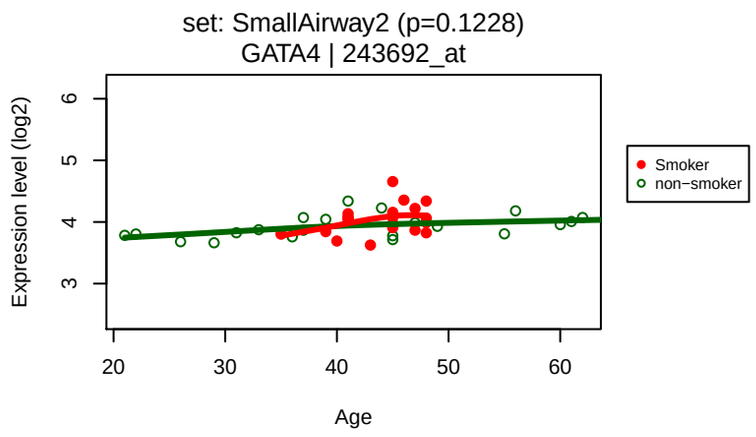
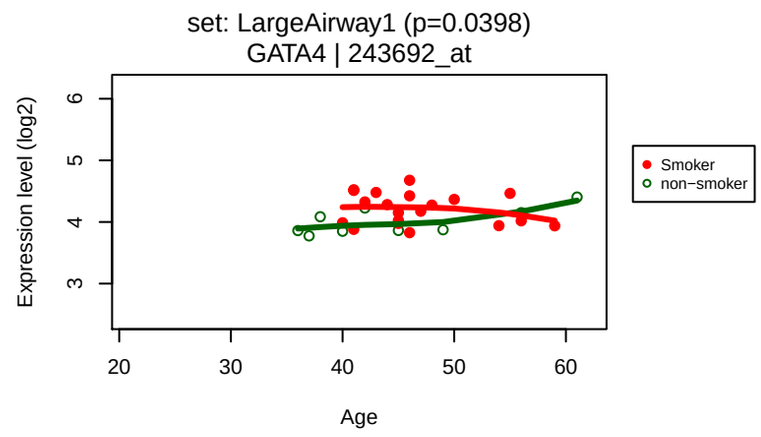

Gene expression of smokers and nonsmokers
gene: GATA4 | 243692_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1405
p=0.1228
p=0.0749
p=0.0398
p=0.3006
p=0.6642
p=0.715
p=0.6712
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

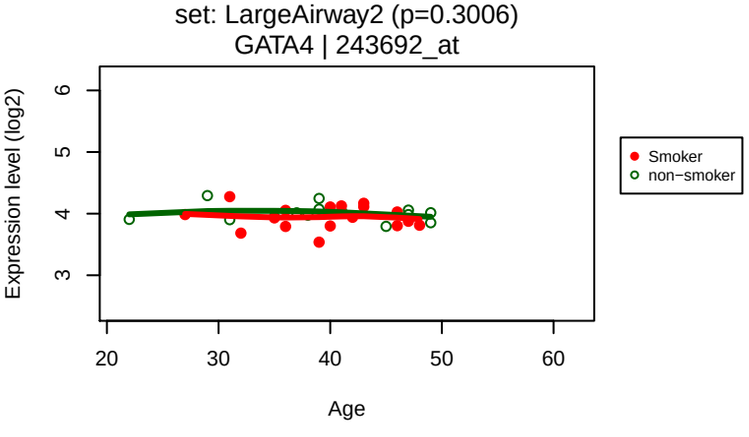

Gene expression ratio (smokers/nonsmokers)
gene: GATA4 | 243692_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1405
p=0.1228
p=0.0749
p=0.0398
p=0.3006
p=0.6642
p=0.715
p=0.6712
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

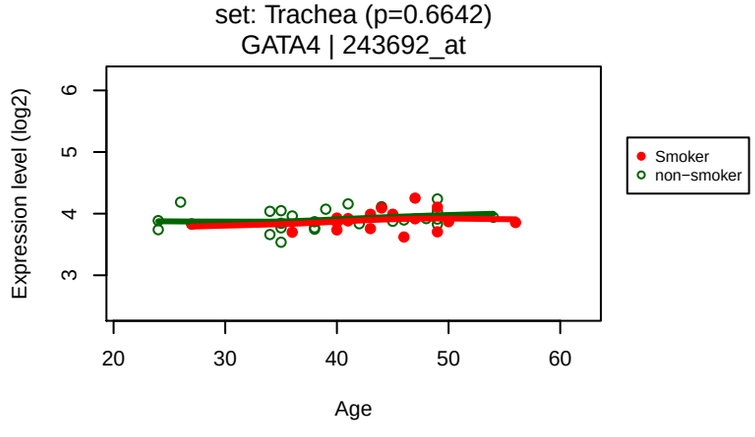

Gene expression ratio in Males
gene: GATA4 | 243692_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1942
p=0.762
p=0.8154
p=0.4164
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATA4 | 243692_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6753
p=0.166
p=0.146
p=0.4039
SmallAirway2
LargeAirway2
Trachea
Macrophage2

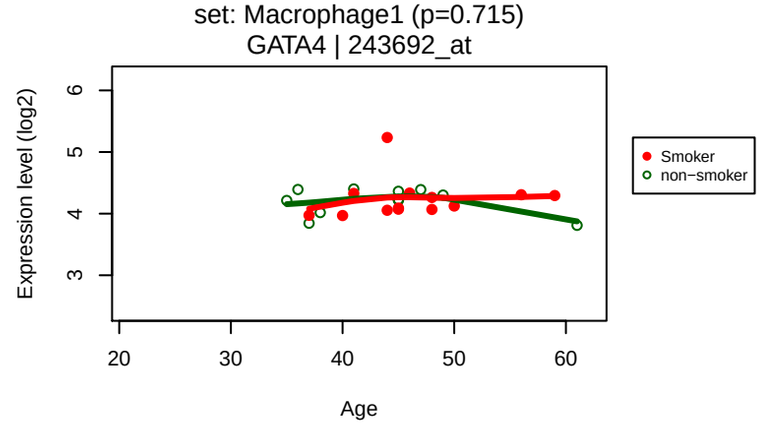

set: SmallAirway1 (p=0.1405)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker

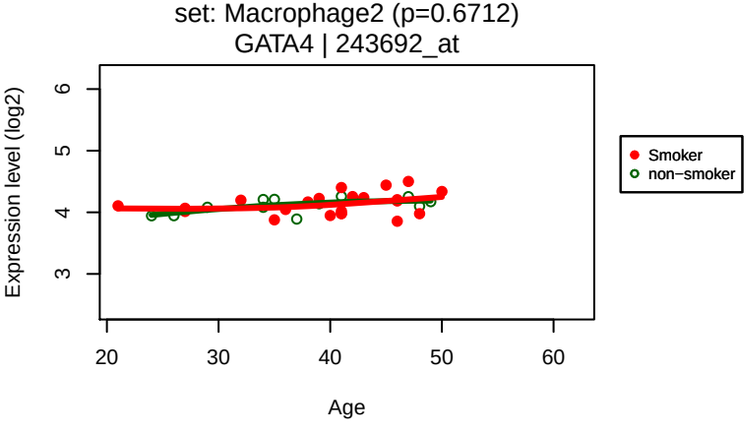

set: SmallAirway2 (p=0.1228)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker

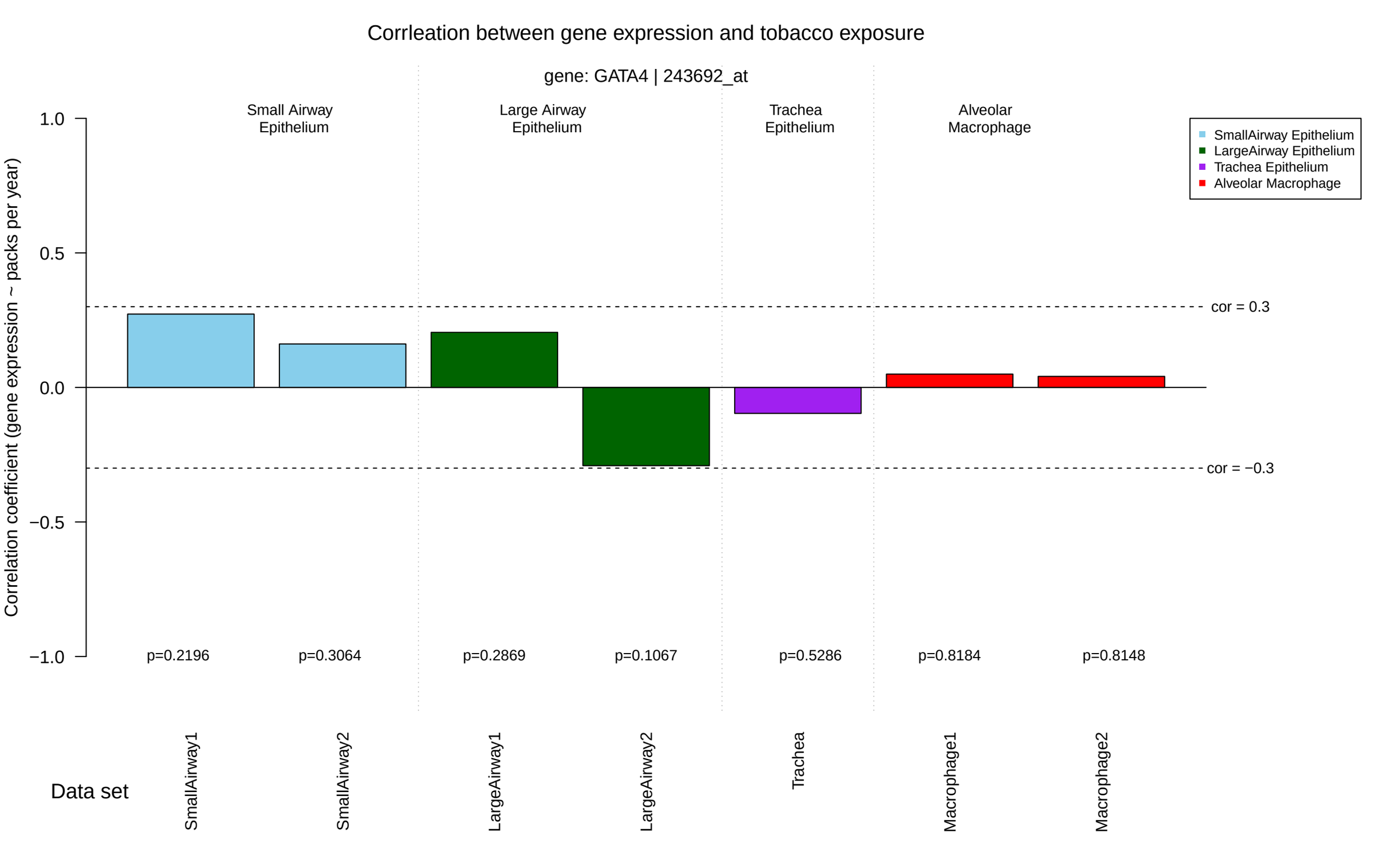

set: LargeAirway1 (p=0.0398)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3006)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6642)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.715)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6712)
GATA4 | 243692_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATA4 | 243692_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2196
p=0.3064
p=0.2869
p=0.1067
p=0.5286
p=0.8184
p=0.8148
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2